RBI Document Question Answering with RAG

Evaluation report • Local open-source models • 27 October 2026

Objective and System

Built a retrieval-augmented question-answering pipeline over RBI documents. The corpus contains 402 parsed documents and approximately 14,816 chunks. The system uses BAAI/bge-base-en-v1.5 embeddings, FAISS vector search, Ollama with Llama 3.1 8B for answer generation, and an optional BAAI/bge-reranker-base cross-encoder for reranking retrieved candidates.

Measured Retrieval Results

Metric	Baseline	With reranking	Change
Recall@5	76.92%	96.15%	+19.23 pp
Recall@10	88.46%	96.15%	+7.69 pp
MRR@10	64.80%	86.22%	+21.42 pp

Both runs used 40 evaluation questions; retrieval metrics are computed over the 26 questions with gold document labels. Recall measures whether a relevant document appears in the top-k retrieved results; MRR@10 rewards relevant documents appearing nearer the top.

Improvements Implemented

1. Grounded prompting: Updated the generation prompt to require answers based only on provided context, avoid unsupported claims, handle ambiguity, and return a fixed refusal when context is insufficient. This prompt change has not yet been isolated in a separate measured answer-quality comparison.
2. Cross-encoder reranking: Retrieves an expanded set of FAISS candidates and reranks query–chunk pairs with BAAI/bge-reranker-base. This change improved all three measured retrieval metrics in the evaluation.

Quality Checks and Limitations

Automated tests: 21 passed, 0 failed. **Baseline full-RAG evaluation:** exact-substring correctness 3.85%; abstention recall on 8 unanswerable questions 0%. These are warning signals, not comprehensive semantic judgments: exact-substring matching is a limited proxy and answers need manual review. The reranking run used retrieval-only mode, so its latency is not directly comparable with the baseline's full-generation latency. Retrieval gains do not by themselves establish answer correctness.

Conclusion and Next Steps

Cross-encoder reranking raised Recall@5 to 96.15% and MRR@10 to 86.22% on the labeled subset. Next, manually review generated answers and citations, improve abstention behavior, record a separate full-RAG evaluation for prompt changes if time permits, then validate the README/demo instructions and tag the final repository as v1.0.